SOAR Playbook Development Documentation

1. Overview

The SOAR playbook was developed to automate incident response for phishing-related alerts detected in the SIEM environment. The objective is to reduce response time, improve accuracy, and standardize actions taken by SOC analysts. The playbook simulates automated workflows including detection, validation, containment, and escalation.

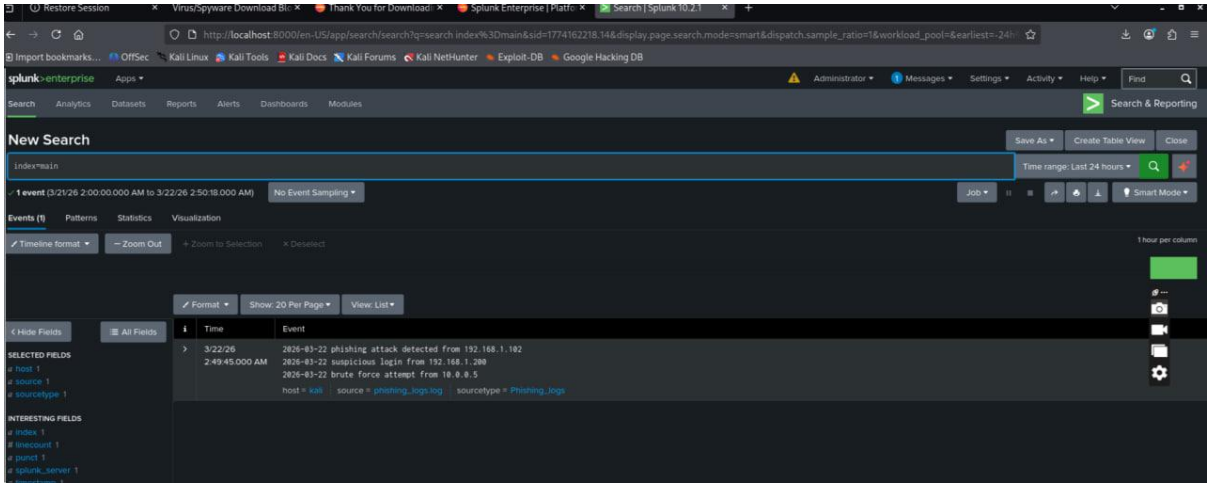
2. Tools Used

Splunk Enterprise – Log monitoring and alert generation
TheHive – Case management and escalation
CrowdSec (simulated via firewall rules) – Threat blocking
Google Docs – Documentation

3. Playbook Workflow

Step 1: Alert Trigger

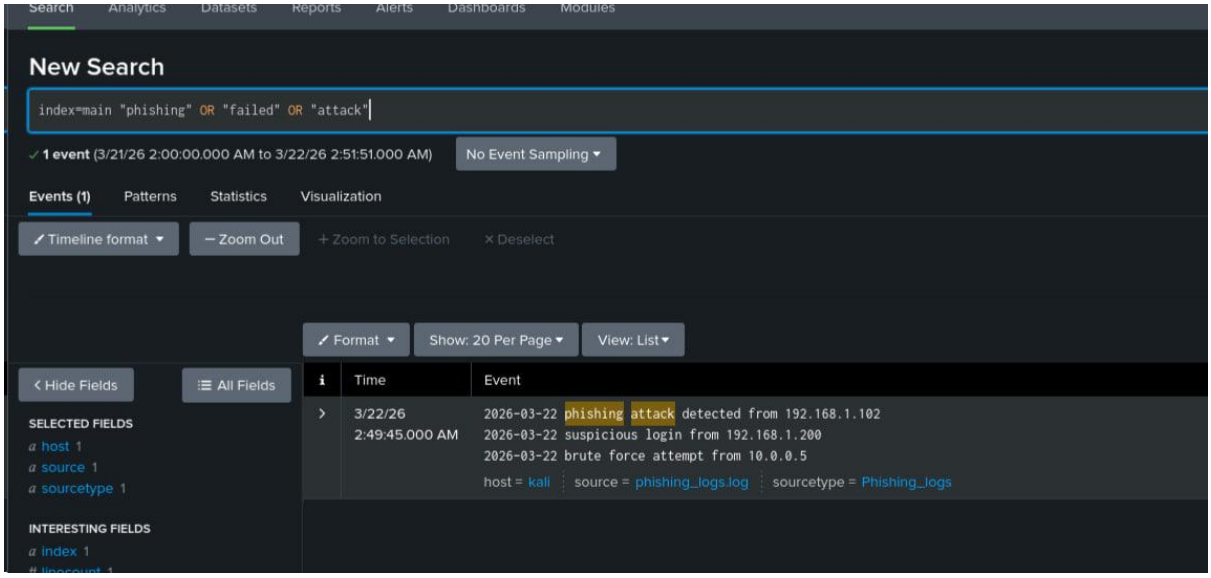
Splunk detects suspicious activity from ingested logs
Example condition:
index=main "phishing" OR "failed login"
Alert triggers when results > 0



Step 2: IP Reputation Check

3. Extract suspicious IP from logs

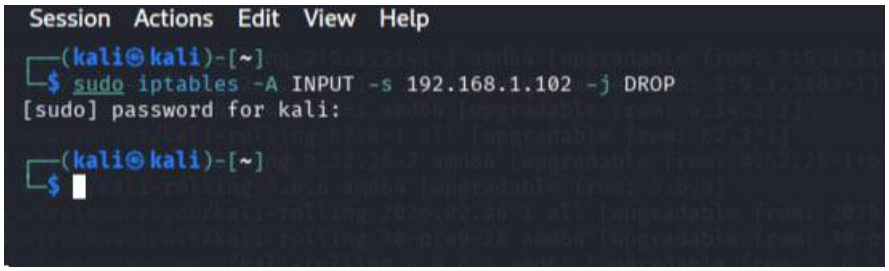
4. Validate using threat intelligence (simulated/manual check)
5. If IP is malicious → proceed to containment



Step 3: Automated Response (Containment)
Block attacker IP using firewall rule:
Bash

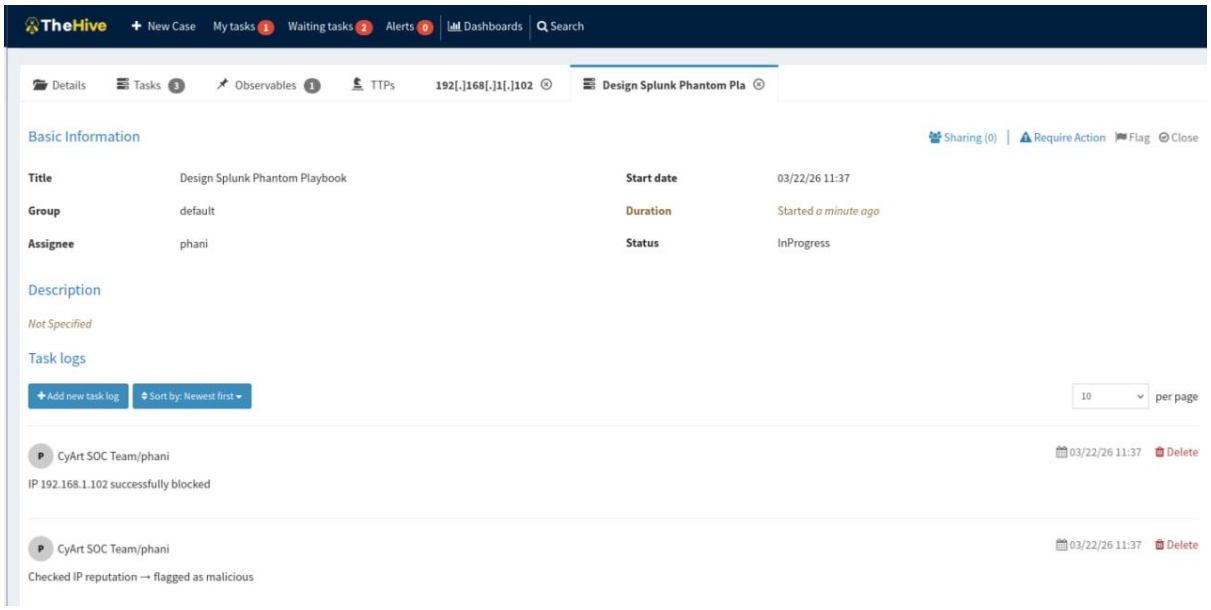
```
sudo iptables -A INPUT -s 192.168.1.102 -j DROP
```

Prevent further communication between attacker and target



Step 4: Case Creation in TheHive
Create a new case in TheHive
Include:

- Alert details
- Source IP
- Description of attack
- Add observables (IP, logs)

- 

Step 5: Task Assignment

- Create investigation tasks:
- Initial Triage
- Log Analysis
- Containment Verification
- Reporting

> CyArt SOC Team/phani
>
> IP 192.168.1.102 successfully blocked
>
> CyArt SOC Team/phani
>
> Checked IP reputation → flagged as malicious

Step 6: Verification

Confirm block using:

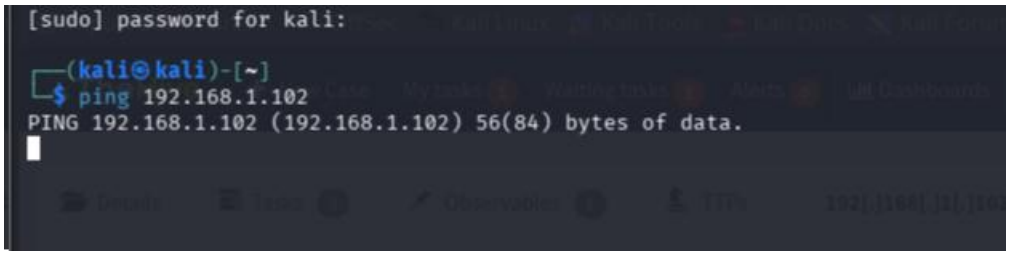
ping 192.168.1.102

No response confirms successful containment

Step 7: Escalation

Escalate case to Tier 2 for deeper analysis

Provide summary and collected evidence



Step 8: Case Closure

Mark case as True Positive

Document resolution and lessons learned

4. Playbook Execution Results

Playbook Step	Status	Notes
Alert Trigger	Success	Splunk detected phishing activity
IP Check	Success	IP identified as suspicious
Block IP	Success	Firewall rule applied
Case Creation	Success	TheHive case created
Verification	Success	Ping test failed (blocked)

5. Playbook Summary (50 Words)

This SOAR playbook automates phishing incident response by detecting threats in Splunk, validating malicious IPs, blocking them via firewall rules, and creating cases in TheHive. It ensures rapid containment, structured investigation, and efficient escalation, significantly reducing manual effort and improving SOC response consistency.